ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** lower temperature	Heat flows downhill in temperature.
2	**(a)** pass on heat to neighbours but stay in position	They vibrate.
3	**(a)** more	They absorb more radiation.
4	**(b)** conduction, convection, radiation	Standard modes.
5	**(a)** moving particles of the fluid	Moving fluid.
6	**(c)** light-coloured clothes	They reflect heat.
7	**(c)** conduction	Heat passes from particle to particle.
8	**(b)** radiation	Waves can travel in vacuum.
9	**(a)** woollen clothes	Wool traps air.
10	**(c)** convection	Hot fluid rises and cold falls.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Conduction and convection (1); Radiation (1).	1+1
12	They reflect more heat (1); Keep us cool (1).	1+1
13	Metals conduct heat (1); Food cooks faster (1).	1+1
14	Transfer of heat through a solid (1); Spoon in hot tea (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fix pins with wax on a rod (1); Heat one end (1); Pins fall one by one (1).	1+1+1
16	Hot air rises (1); Cool air sinks (1); Convection spreads it (1).	1+1+1
17	Trap air (1); Insulation (1); Retain body heat (1).	1+1+1
	OR: Water at the bottom gets hot (1); Hot water rises (1); Cool water sinks to replace it (1).	1+1+1
18	Metal conducts heat quickly from hand (1); Wood is a poor conductor (1); We feel the heat loss (1).	1+1+1
	OR: Cloth is a poor conductor (1); Prevents burns (1); Heat passes slowly (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Conduction: spoon in tea (1); Convection: boiling water (1); Radiation: Sun (1); Conduction needs solid (1); Radiation needs no medium (1).	1+1+1+1+1
	OR: Medium needed (1); Direction (1); Particle movement (1); Example (1); Speed (1).	1+1+1+1+1
20	Heat water in a beaker (1); Place crystal at the bottom (1); Coloured streaks rise (1); Move around the beaker (1); Show convection currents (1).	1+1+1+1+1
	OR: Heat conducts through base (1); Water conducts and convects (1); Soup circulates (1); Even cooking (1); Heat reaches food (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** convection currents	Hot water rises.
2	**(d)** traps air, a poor conductor	Trapped air insulates.
3	**(d)** radiation	E.g. from the Sun.
4	**(b)** a metal rod in a flame	Metal rod.
5	**(a)** reducing conduction, convection and radiation	It has a vacuum.
6	**(c)** radiation	Space has no medium.
7	**(a)** a person sitting in sunlight	Direct from the Sun.
8	**(c)** radiation	They reflect heat.
9	**(d)** aluminium	Metals conduct heat.
10	**(d)** day	From sea to land.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wool traps air (1); Air is a poor conductor (1).	1+1
12	Poor conductors (1); Do not burn hands (1).	1+1
13	Heat transfer by movement of fluid (1); Boiling water (1).	1+1
14	Wind from sea to land by day (1); Due to differing heating (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No medium in space (1); By radiation (1); Travels as waves (1).	1+1+1
16	Double-walled glass (1); Vacuum between walls (1); Silvered surfaces and cork (1).	1+1+1
17	Let in sunlight (1); Trap warm air (1); Keep plants warm (1).	1+1+1
	OR: Land heats faster in the day (1); Warm air above land rises (1); Cool air from sea moves in (1).	1+1+1
18	Conductors: copper, iron, aluminium (1); Insulators: wood, plastic, air (1); Use in utensils (1).	1+1+1
	OR: Heat radiates (1); Warm air rises and circulates (1); Convection currents (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day: land hotter (1); Warm air rises over land (1); Cool sea air moves to land (1); Night: reverse (1); Land breeze from land to sea (1).	1+1+1+1+1
	OR: Handles (1); Thermocol boxes (1); Wool clothes (1); Double-glazed windows (1); Wire covers (1).	1+1+1+1+1
20	Light colours reflect (1); Dark colours absorb (1); Cotton in summer (1); Wool in winter (1); Air traps heat (1).	1+1+1+1+1
	OR: Burner heats the air (1); Hot air is lighter (1); It rises (1); Cool air pushed in (1); Convection principle (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
- Thinking that heat from the Sun reaches us by conduction or convection; it is **radiation**, which needs no medium.
- Writing that dark colours are better in summer; **light** colours reflect more radiation.
- Confusing heat with temperature; temperature is a measure of hotness, heat is energy in transfer.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Heat Transfer in Nature** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** conduction and convection	No medium.
2	**(c)** wood	Wood.
3	**(c)** night	From land to sea.
4	**(c)** is a poor conductor	Prevents heat transfer.
5	**(d)** conduction	Along the solid.
6	**(d)** rises	Warm air rises.
7	**(a)** reflector	Reflects.
8	**(b)** convection	Fluid moves.
9	**(c)** air over land is warmer and rises	Cooler air moves in.
10	**(d)** absorber	Absorbs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To stop conduction and convection (1); Silvering stops radiation (1).	1+1
12	Transfer of heat without a medium (1); Heat from the Sun (1).	1+1
13	Wind from land to sea by night (1); Land cools faster (1).	1+1
14	Cool air sinks (1); Convection currents form (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Land cools faster (1); Air above sea is warmer and rises (1); Cool air from land moves to sea (1).	1+1+1
16	Filled with air spaces (1); Air is a poor conductor (1); Body heat stays (1).	1+1+1
17	Metal conducts heat (1); Wooden handle insulates (1); Safe to hold (1).	1+1+1
	OR: White reflects heat (1); Dark absorbs heat (1); Less absorption (1).	1+1+1
18	Hot water rises (1); Whole water gets heated by convection (1); Heating from above only warms the top (1).	1+1+1
	OR: Snow traps air (1); Air insulates (1); Body heat retained (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vacuum stops conduction and convection (1); Silvered walls reflect radiation (1); Cork stopper (1); Glass is a poor conductor (1); Keeps contents hot or cold (1).	1+1+1+1+1
	OR: Moving air takes sweat away (1); Evaporation cools (1); Warm air replaced (1); In winter not needed (1); Convection helps (1).	1+1+1+1+1
20	(a) Wood is an insulator (1); (b) Cold air sinks (1); (c) Reflect heat (1); (d) Trapped air (1); (e) Reduces heat gain (1).	1+1+1+1+1
	OR: Reflector focuses sunlight (1); Black surface absorbs (1); Glass lid traps heat (1); Food cooks (1); No fuel needed (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that heat flows from a cold body to a hot body; heat always flows from **hot to cold**.
- Writing that conduction needs the movement of particles from place to place; in conduction the particles only pass on the heat and **stay in position**.
- Mixing up sea and land breezes: the *sea breeze* blows by day (sea to land), the *land breeze* blows at night (land to sea).
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